

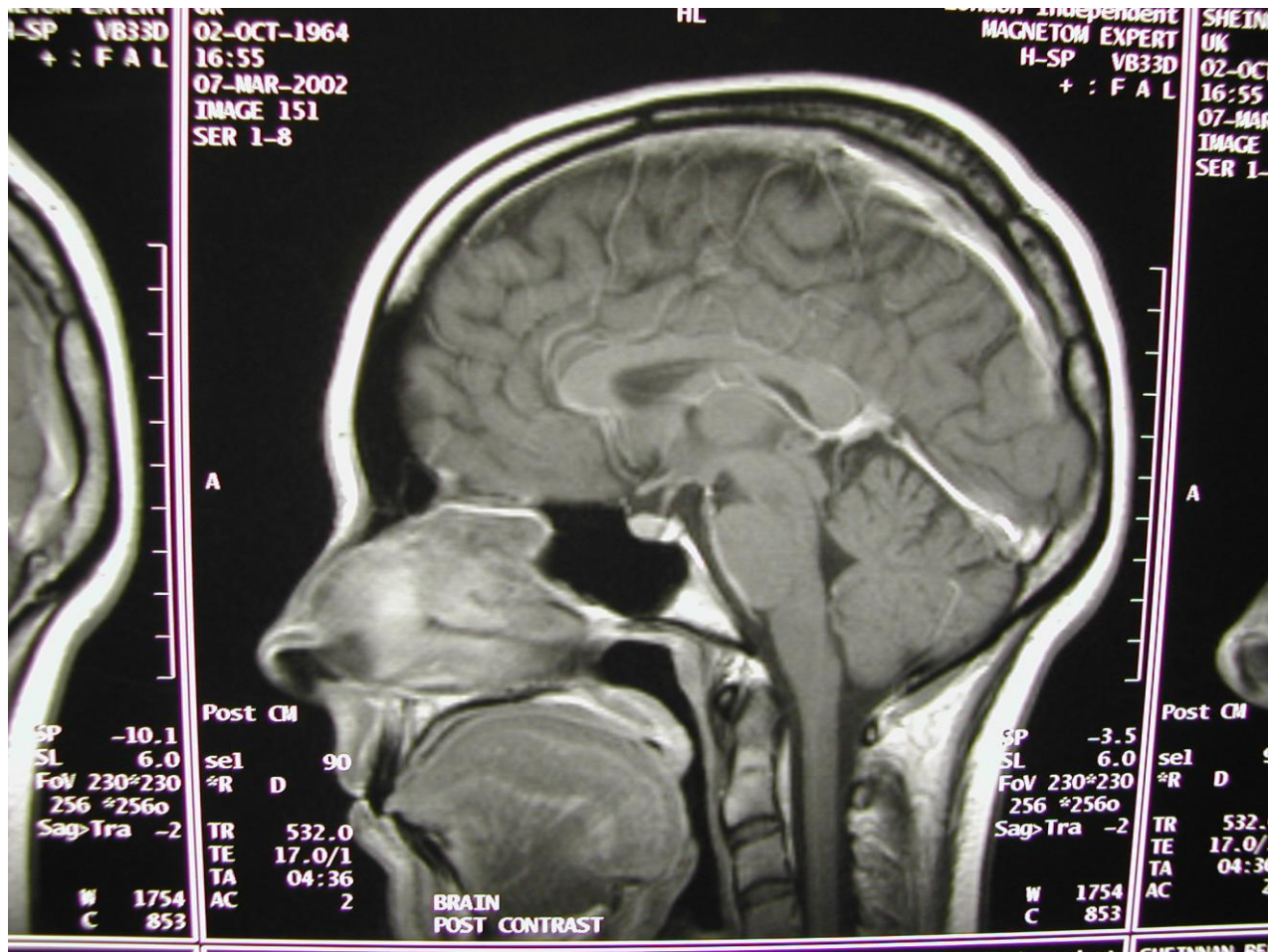
Function and structure of the brain

Professor Michael Swash
Queen Mary University of London
and CMO SwissRe

Anatomy determines function

- Neurons and axons (white matter)
- Nuclei of neurons
- Specialised cortical areas
- Spinal cord sensory and motor pathways
- Cerebellum and basal ganglia
- Sensory modalities; sensient and insensient
- Motor control theory

Normal brain – sagittal section





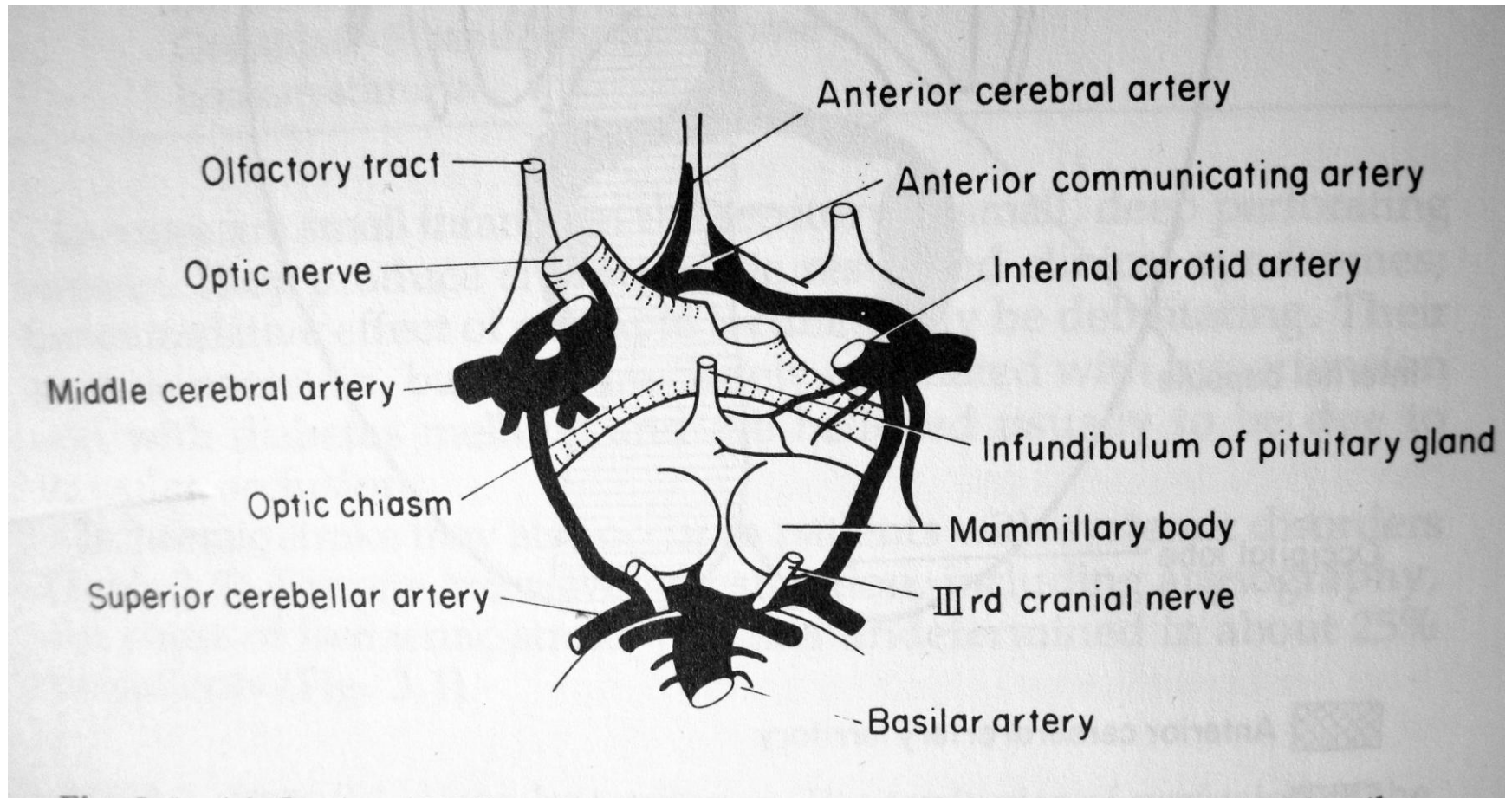
circulation

- Arteries and veins
- Capillary circulation
- Anastomoses and end-arteries
- Cardiac arrest causes loss of consciousness in a few seconds
- Only 5 minute window before brain death

Normal brain and circle of Willis



Circle of Willis and its branches



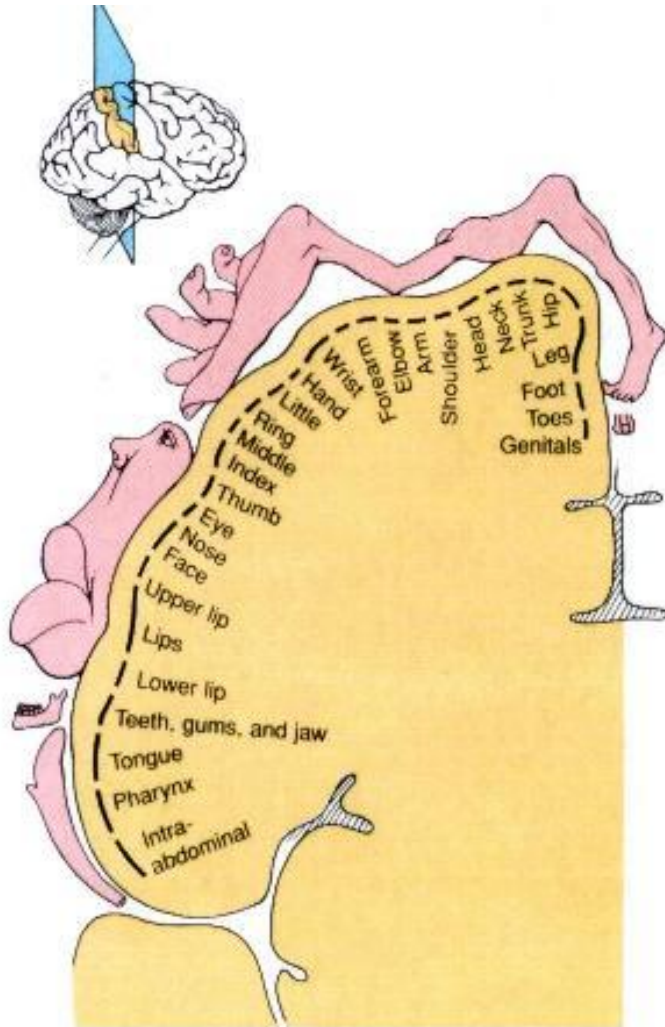
Clinical kit



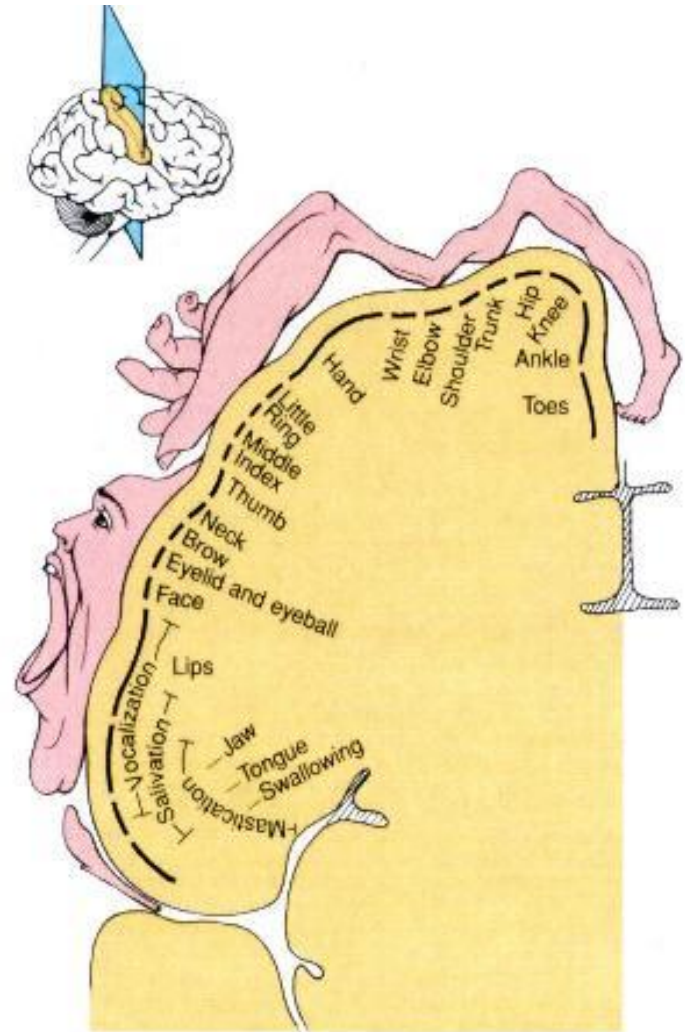
Cortical specialisation

- The cortex is not holistic
- Other parts of the brain cannot take over functions from damaged areas
- Specialised brain areas are connected with other parts in a logical and essentially useful manner

Motor & Sensory Homunculi



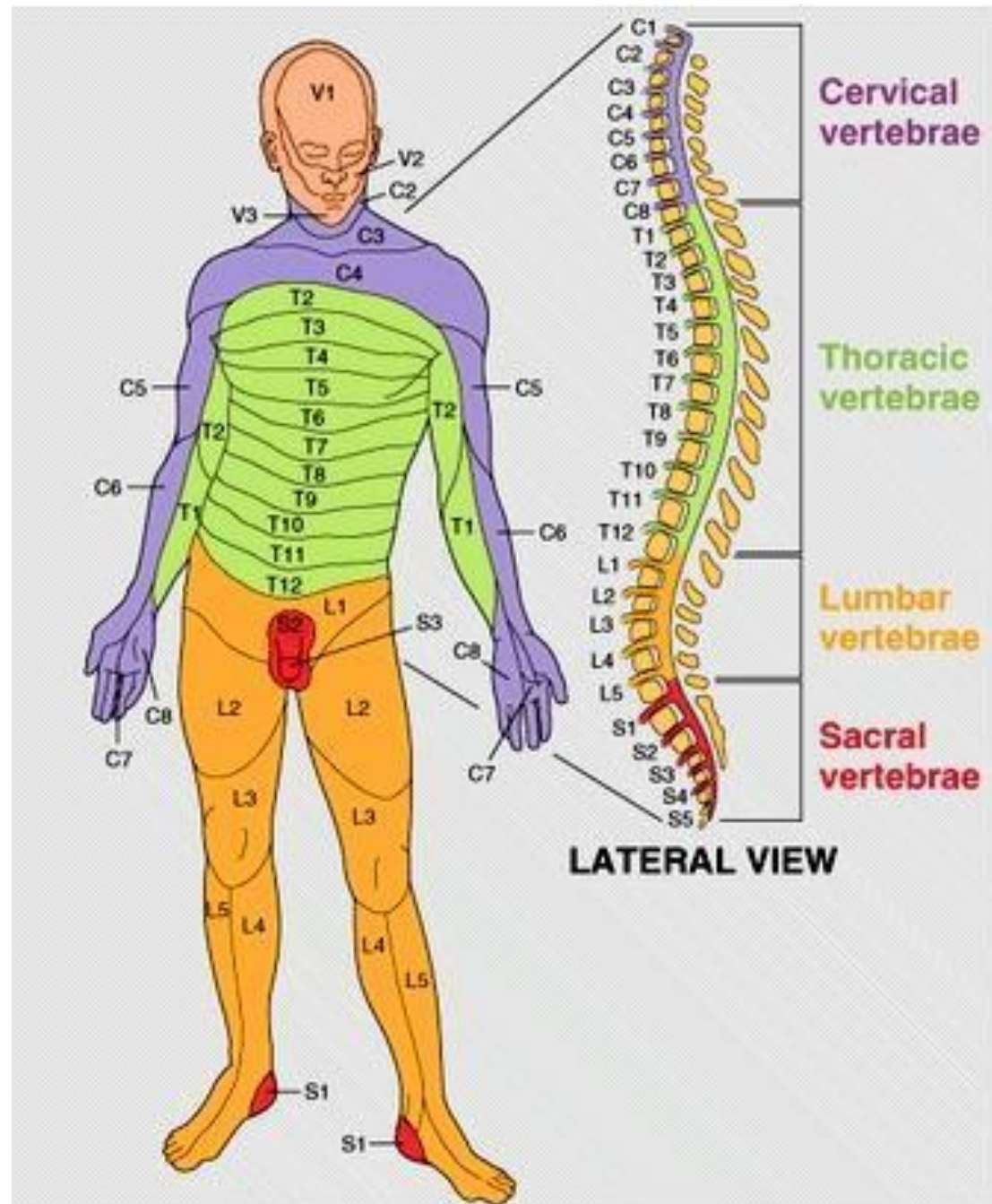
(a) Somatosensory cortex in right cerebral hemisphere



(b) Motor cortex in right cerebral hemisphere

Dermatomes

– The Big Picture



Radicular Pain - Arm

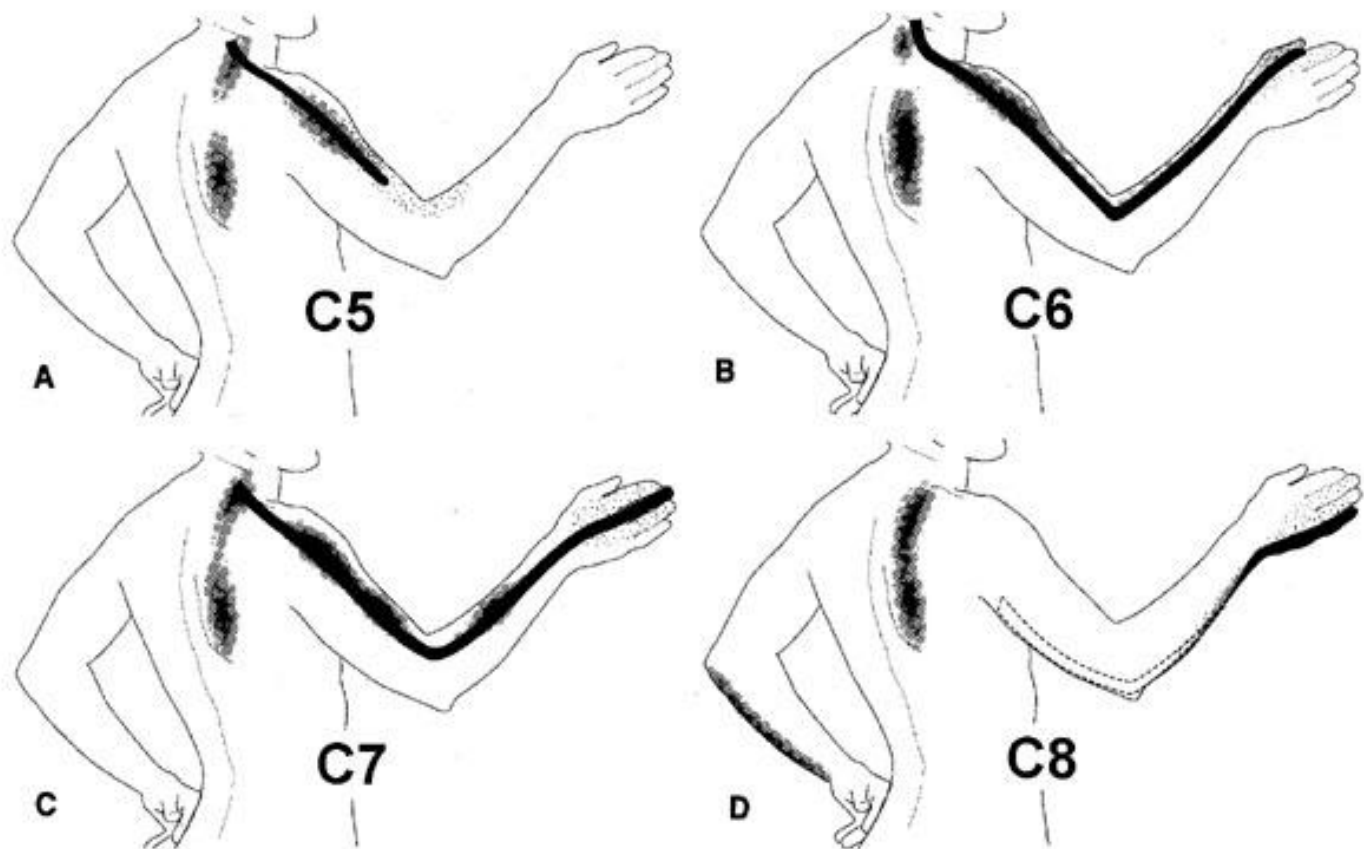


Fig. 67. Representation of the anterolateral aspect of the neck, shoulder, and upper limb. The thick black line represents the sharp, radiating pain, which often has a dermatomal distribution. Interrupted lines indicate sharp pain with a C8 radiculopathy, which is on the inner aspects of the arm and forearm. The diffuse gray areas represent the poorly localized dull ache. A dull ache medial to the shoulder blade is a common complaint in all cervical radiculopathies and is of no localizing value. The area covered by small dots indicates the location of paraesthesiae and sensory impairment. **A.** C5 radiculopathy. **B.** C6 radiculopathy. **C.** C7 radiculopathy. **D.** C8 radiculopathy.

Patterns of disease presentation and progression

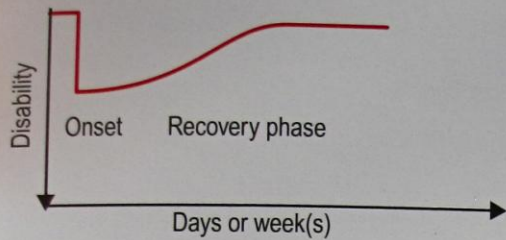


Fig. 51 Tempo of stroke illness; sudden onset and gradual functional improvement or even recovery.

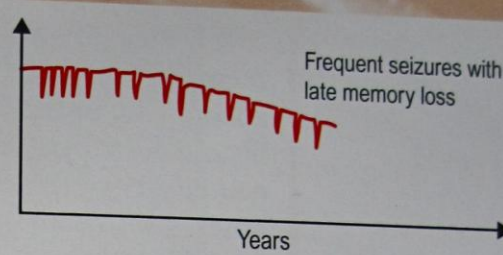


Fig. 52 Tempo of epilepsy: recurrent stereotyped seizure events; with generalized epilepsy there may be the late onset of memory and personality deficit.

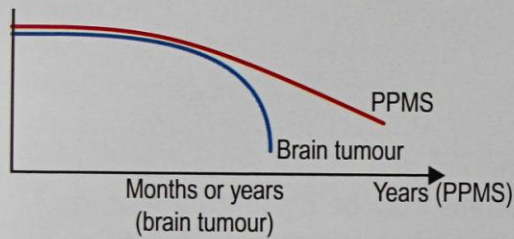


Fig. 53 Tumour or primary progressive multiple sclerosis (PPMS). Intractable gradual progression, accelerating in the case of tumour.

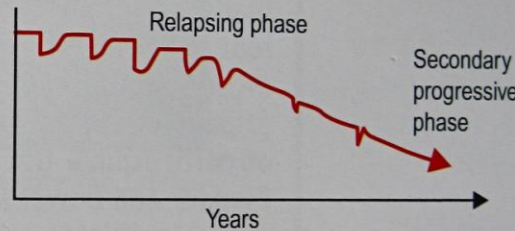


Fig. 54 Relapsing-remitting multiple sclerosis, with later onset secondary progression.

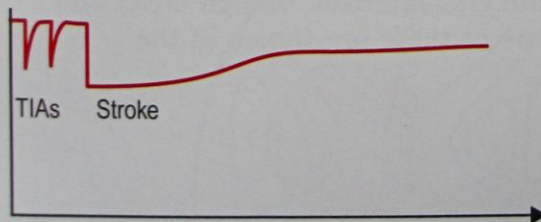


Fig. 55 TIA and stroke; recurrent brief episodes may be followed by a disabling stroke.

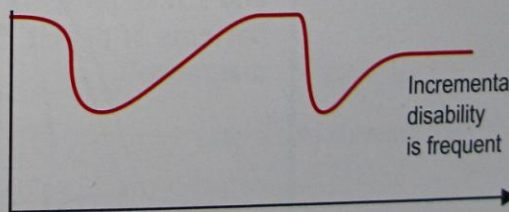


Fig. 56 Relapsing neuropathy (CIDP); phases of progression, often rapid, with gradual recovery.

Patterns of neurological disease in various conditions:

Stroke

Epilepsy

PPMS

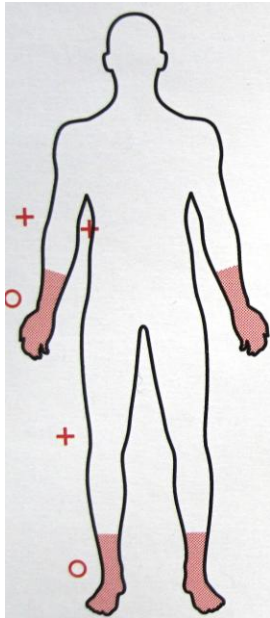
Tumour

RRMS

TIA & stroke

CIDP

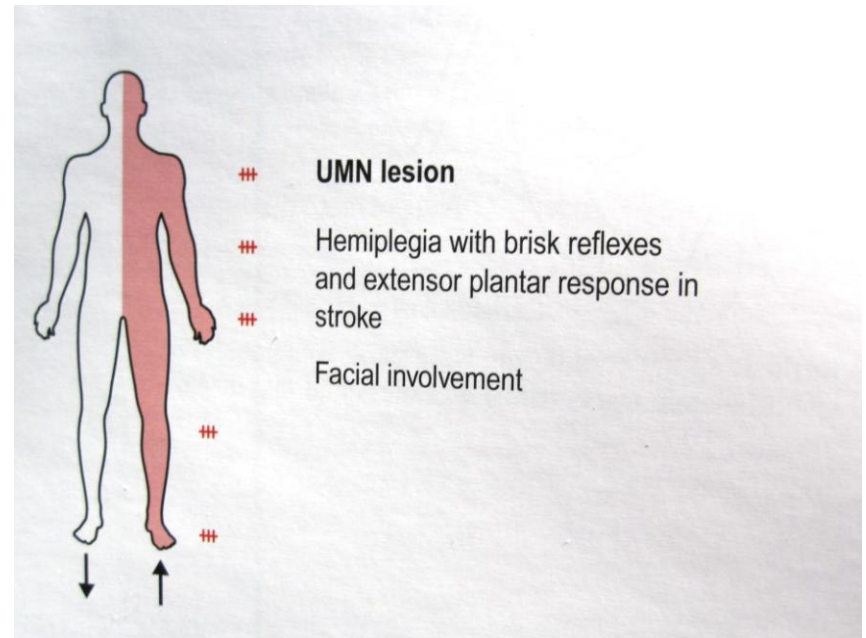
Peripheral neuropathy and brain lesions



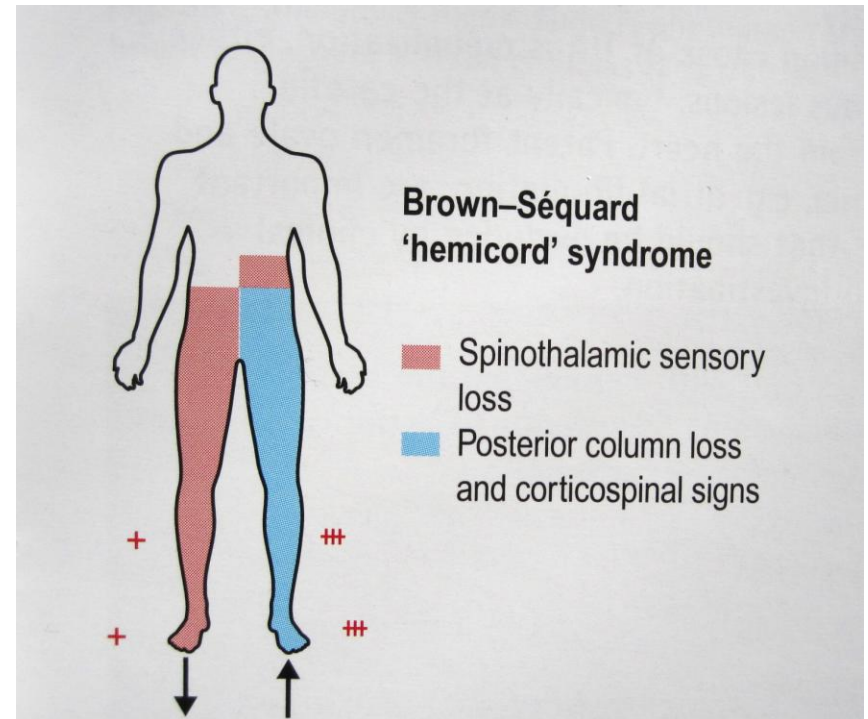
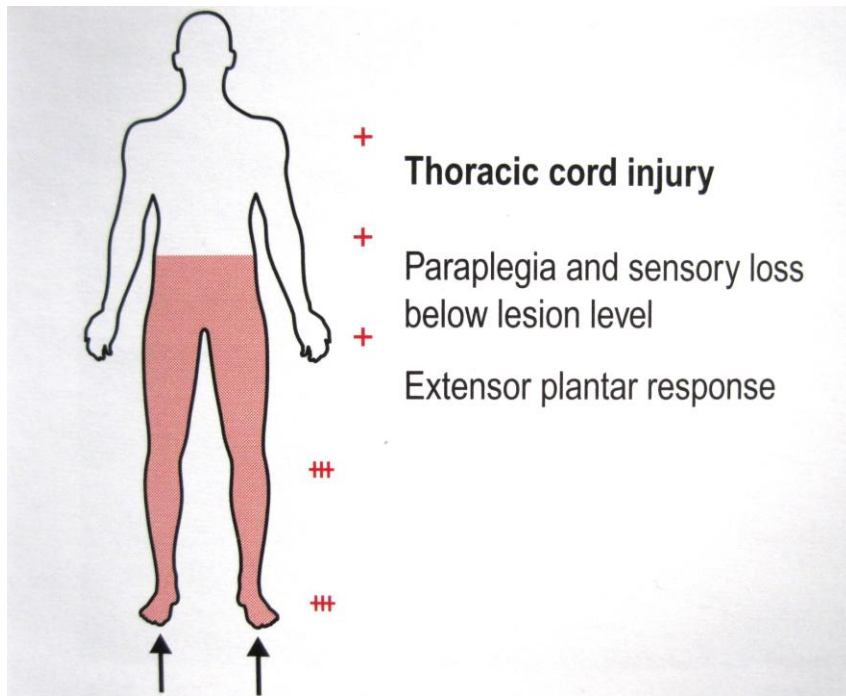
Peripheral neuropathy

Axonal-type peripheral neuropathy with distal weakness and sensory loss

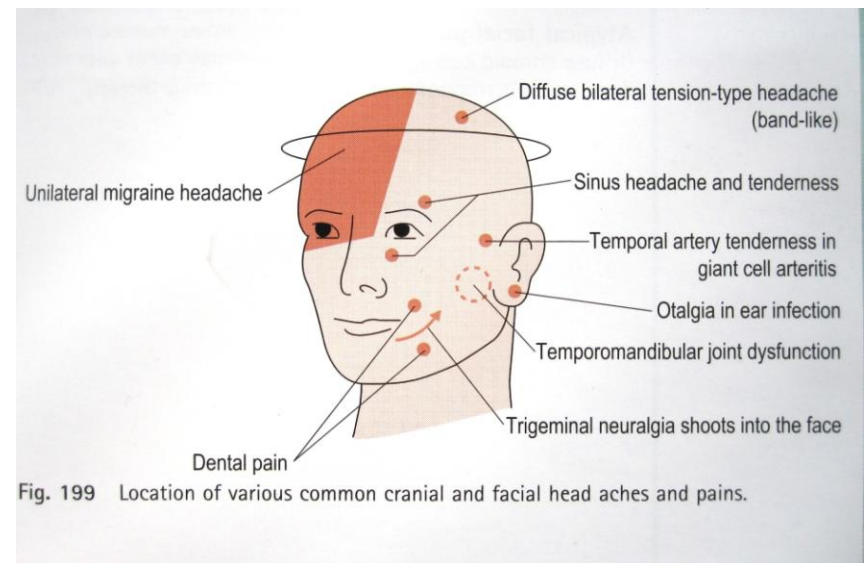
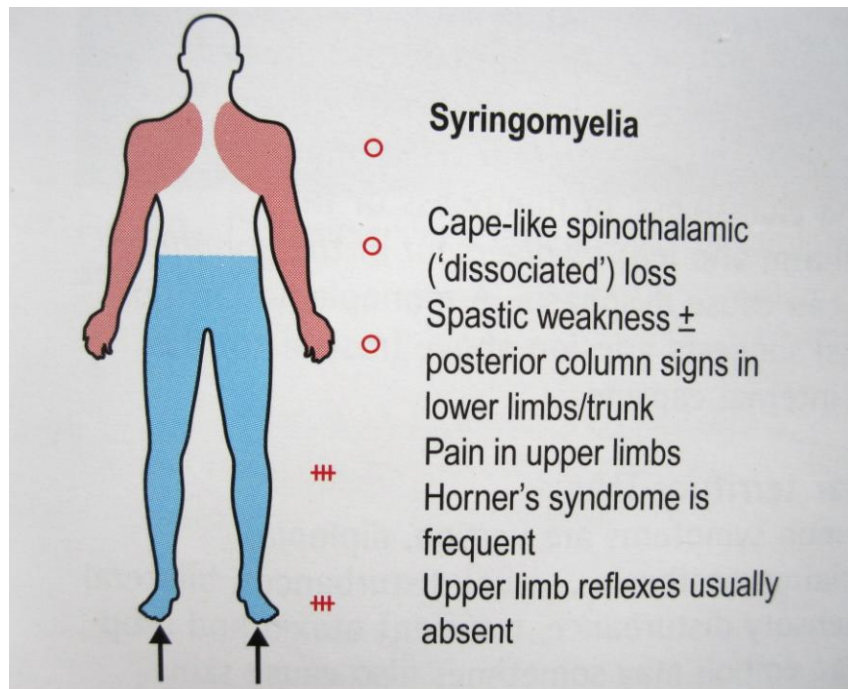
Absent distal reflexes



Spinal cord syndromes



Syringomyelia and headache



Spinal muscular atrophy

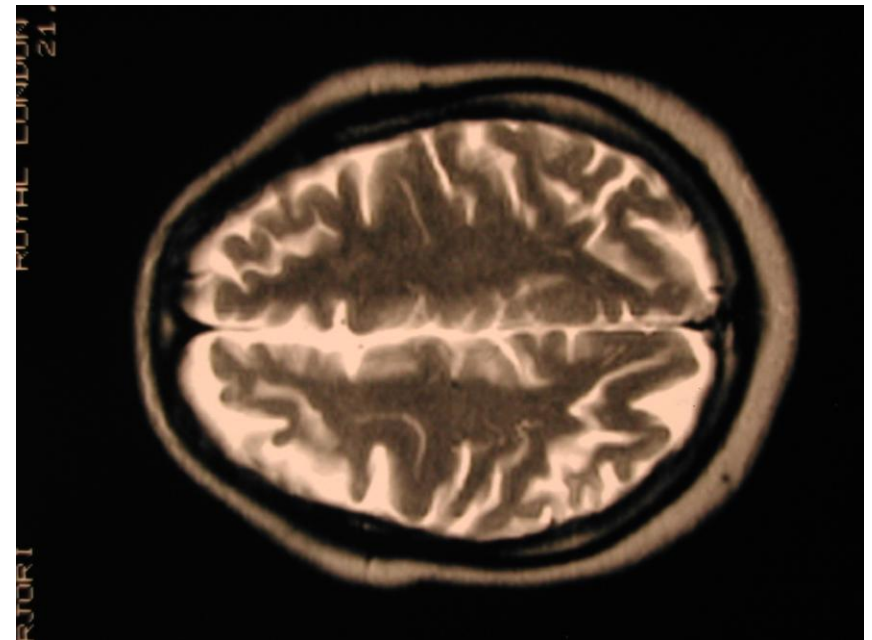
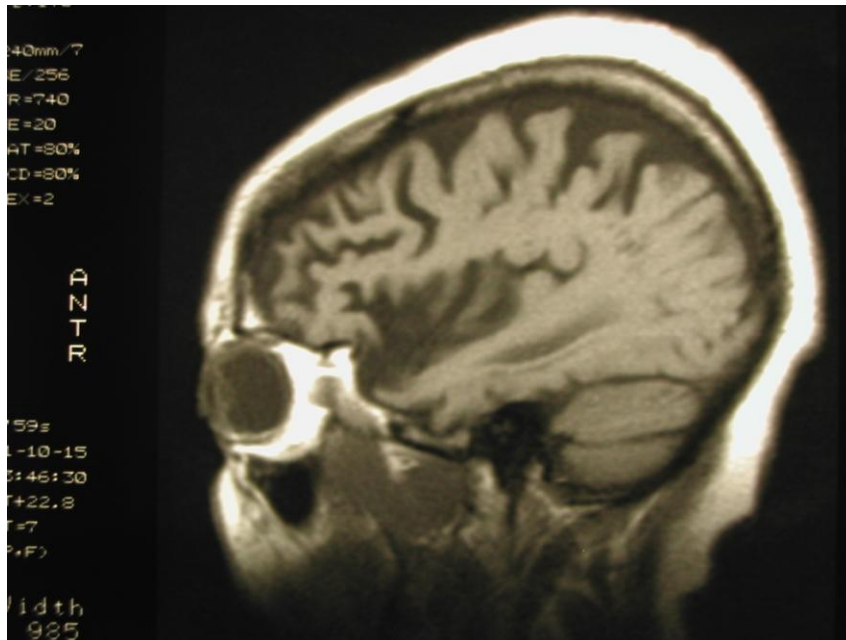


- Recessively inherited neuromuscular disease
- Progressive at onset
- Childhood or adolescent types
- Variable severity
- Life expectancy can be normal

Musician's dystonia



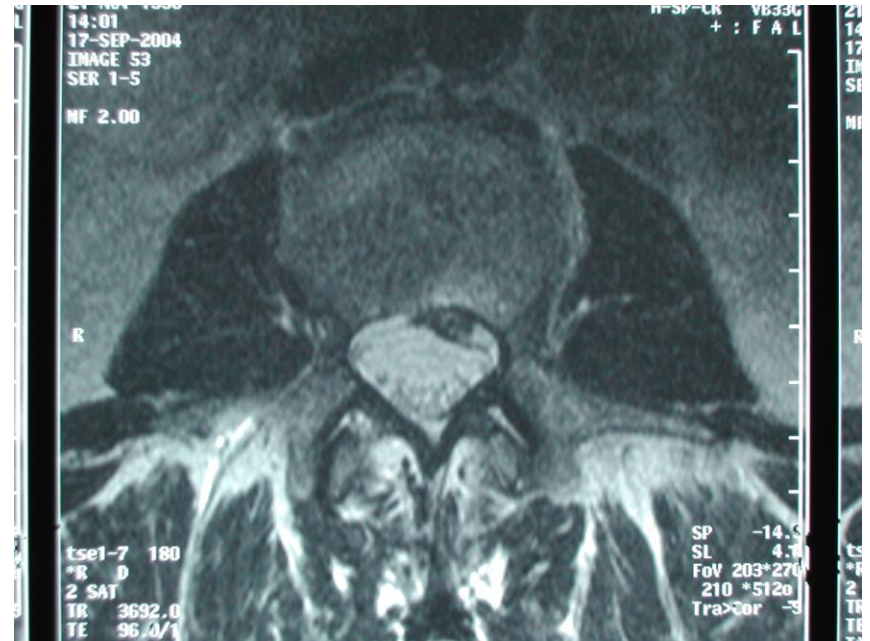
Dementia and brain atrophy



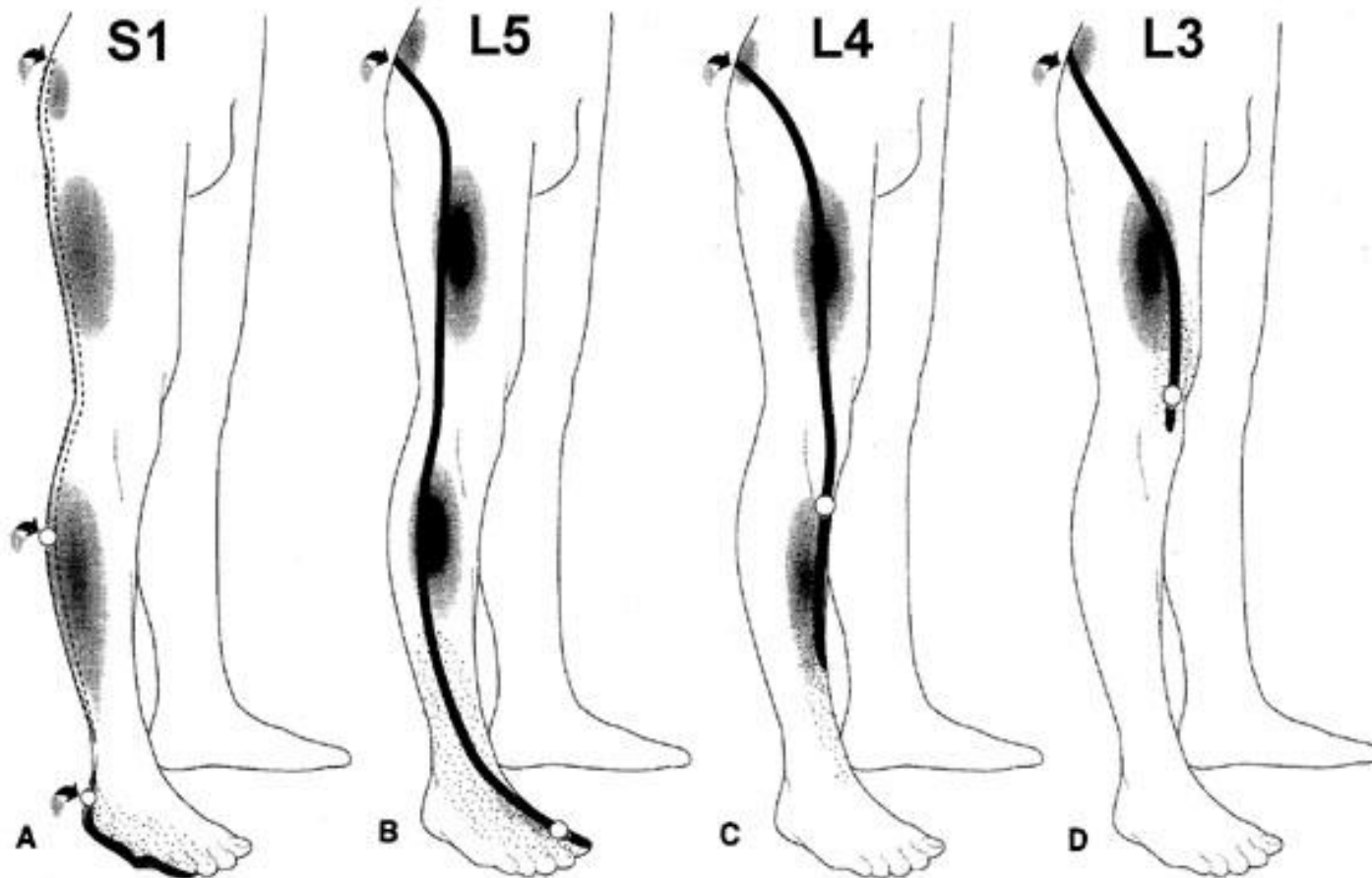
Cerebellar atrophy; a group of progressive genetic disorders



Lumbosacral disk protrusion



Radicular Pain - Leg



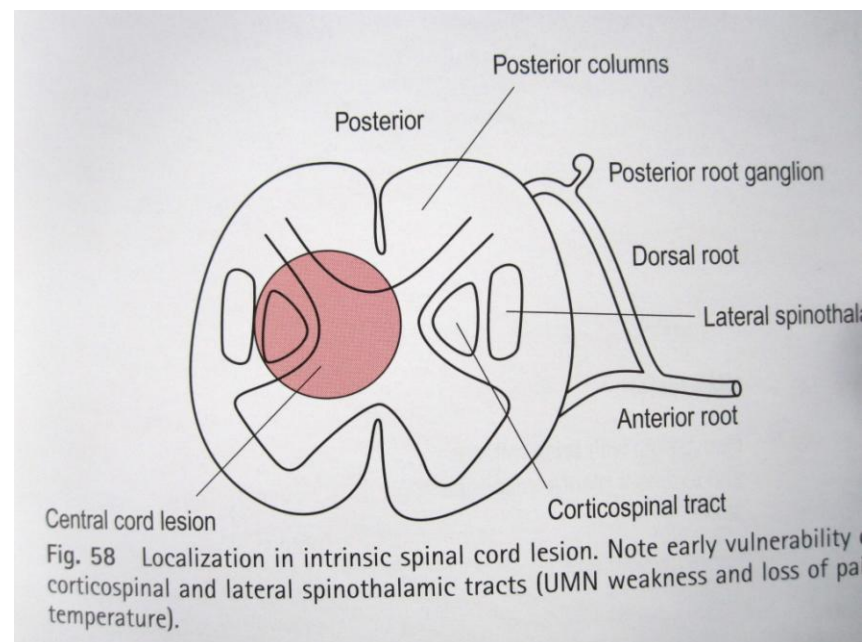
Anterolateral view of the lower extremity. The black thick line represents the sharp, radiating pain, which often has a dermatomal distribution. The sharp radiating pain in S1 radiculopathy is indicated by interrupted lines. It tends to be in the center of the posterior thigh and calf. The diffuse gray areas represent the poorly localized dull ache. The circles indicate areas where pain may concentrate. The area covered by small dots indicates the location of paraesthesiae and sensory impairment. **A.** S1 radiculopathy. **B.** L5 radiculopathy. **C.** L4 radiculopathy. **D.** L3 radiculopathy.

Stroke

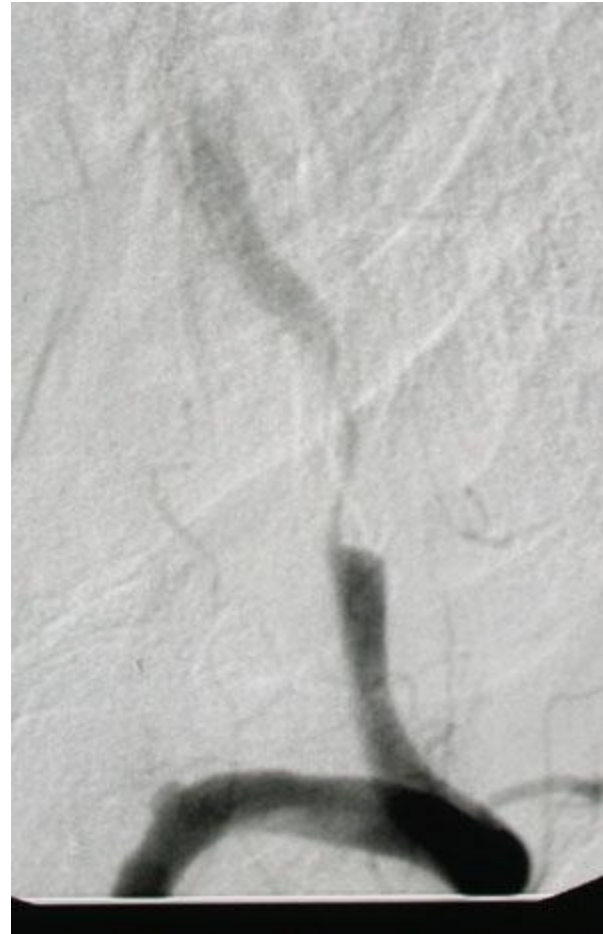
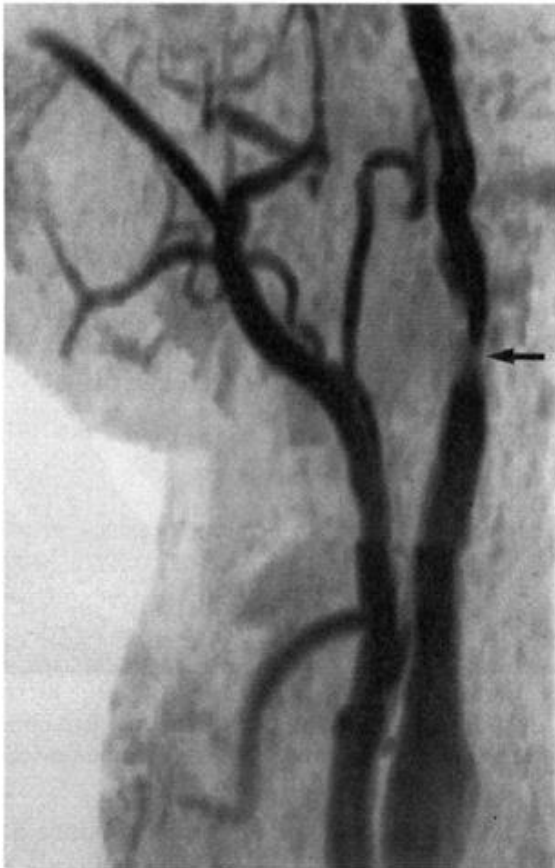
CT Scan – Cerebral Infarct



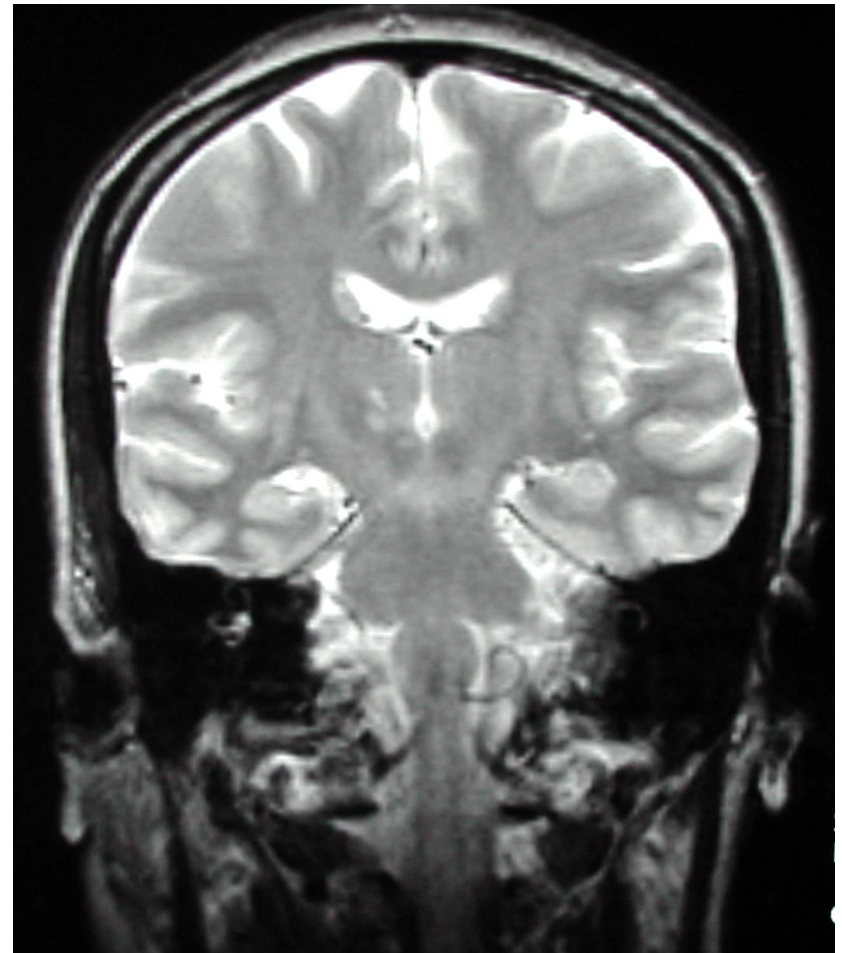
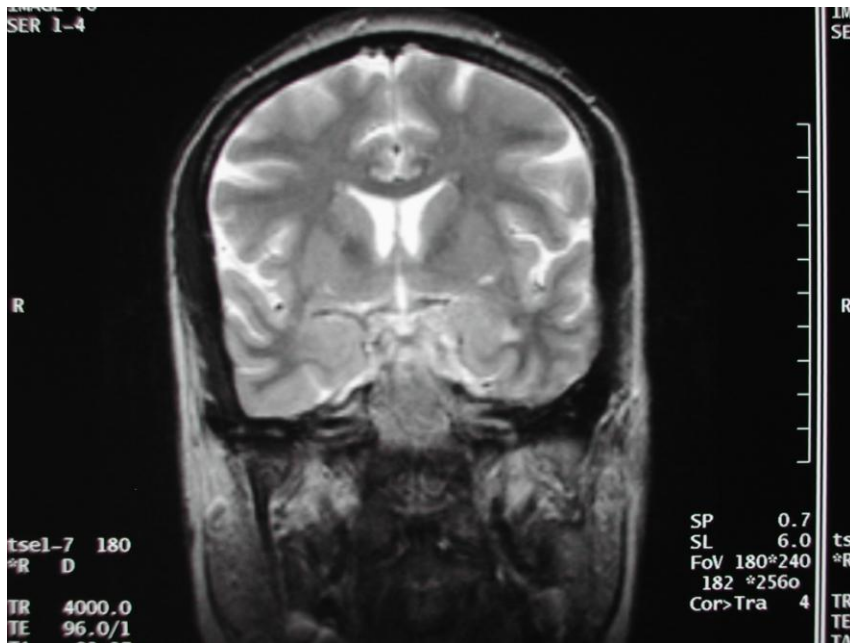
Spinal cord disease



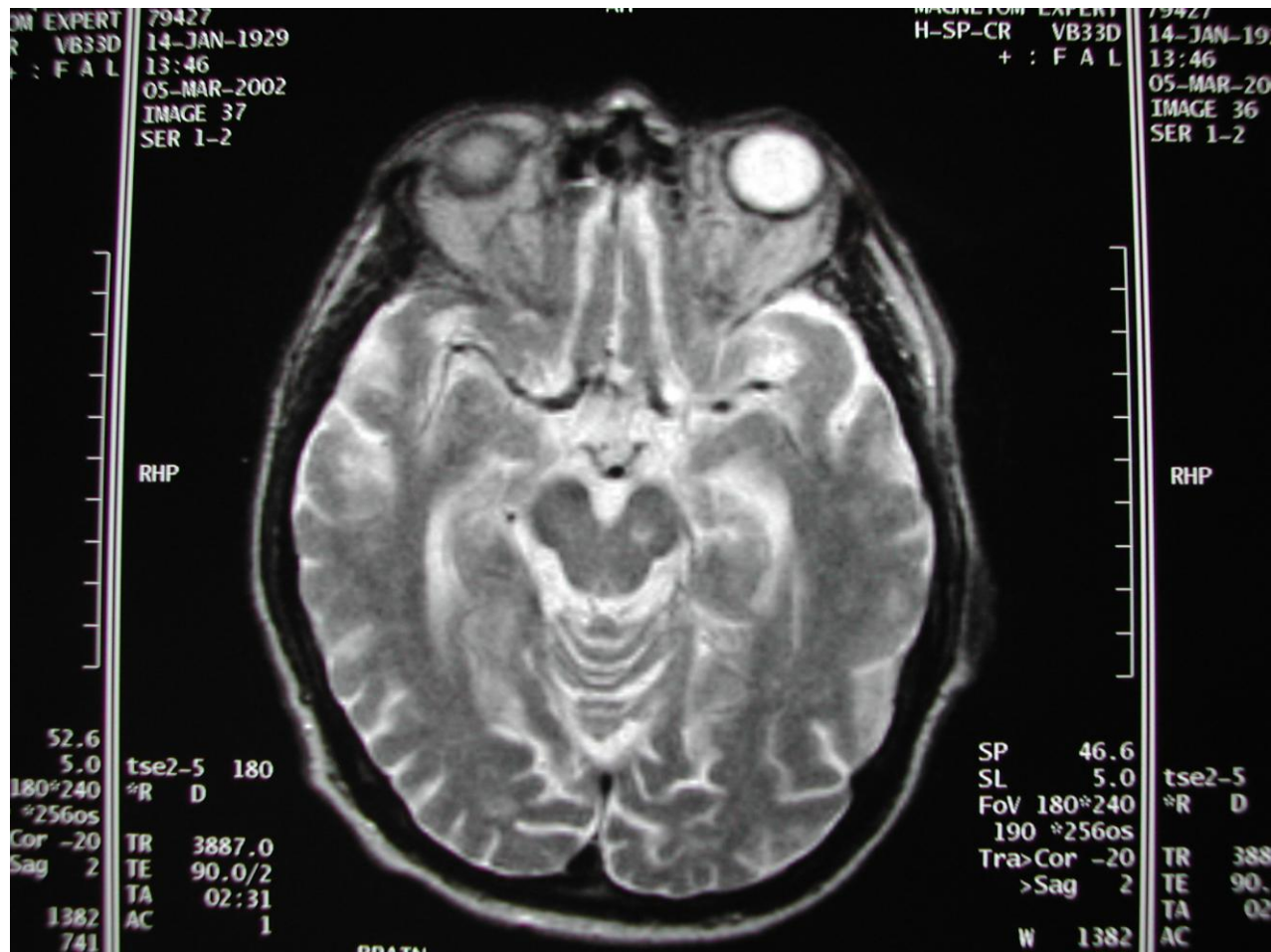
Atheroma or dissection



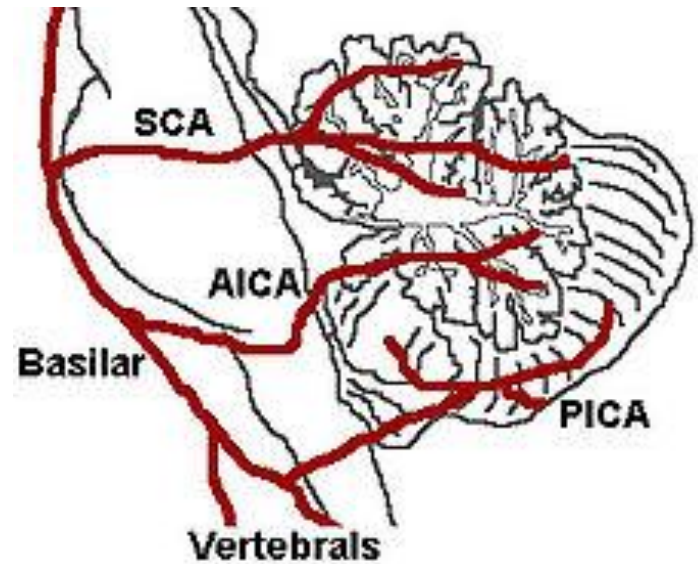
Lacunar infarction in deep white matter or basal ganglia



Lacunar pontine infarction



Left Cerebellar Stroke (PICA)



Cerebellar Haematoma of hypertensive aetiology

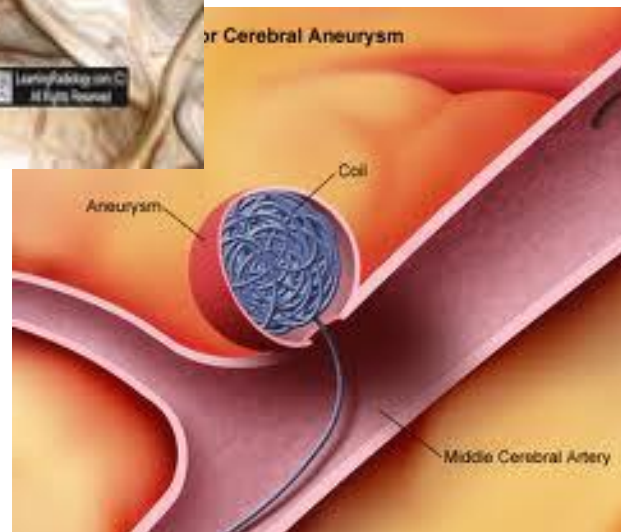


Subarachnoid haemorrhage

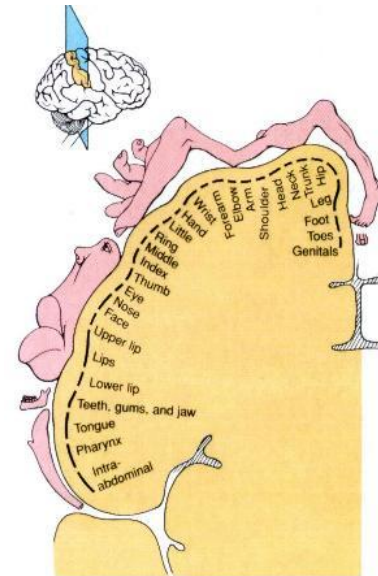
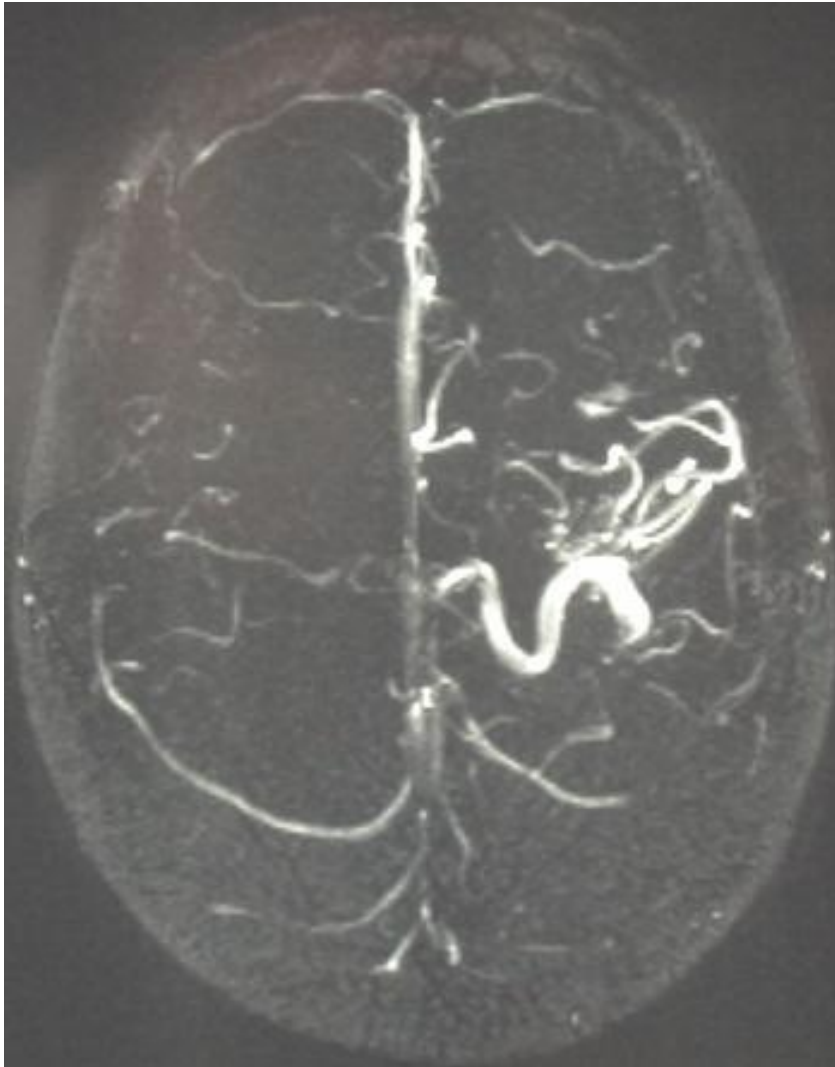
Aneurysmal subarachnoid haemorrhage



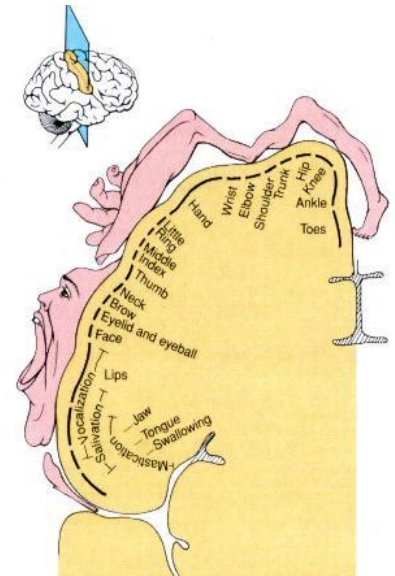
Aneurysms & Subarachnoid Haemorrhage



AVM – Left Fronto-Parietal



(a) Somatosensory cortex in right cerebral hemisphere



(b) Motor cortex in right cerebral hemisphere

syringomyelia



- Central CSF-filled cavity extending through part of cord
- Usually in cervical region, may involve brainstem; painful and progressive
- weak legs and cape-like pin prick loss

Syringomyelia – a progressive spinal cord malformation

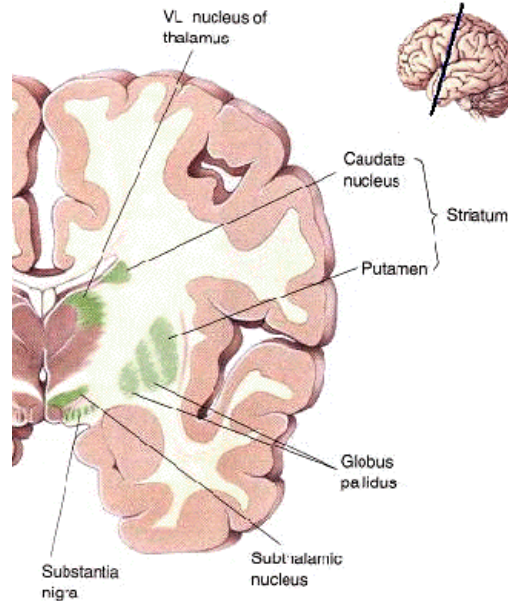
Due to “water-hammer” effect on central cord structure

Parkinson's disease and other degenerative disorders

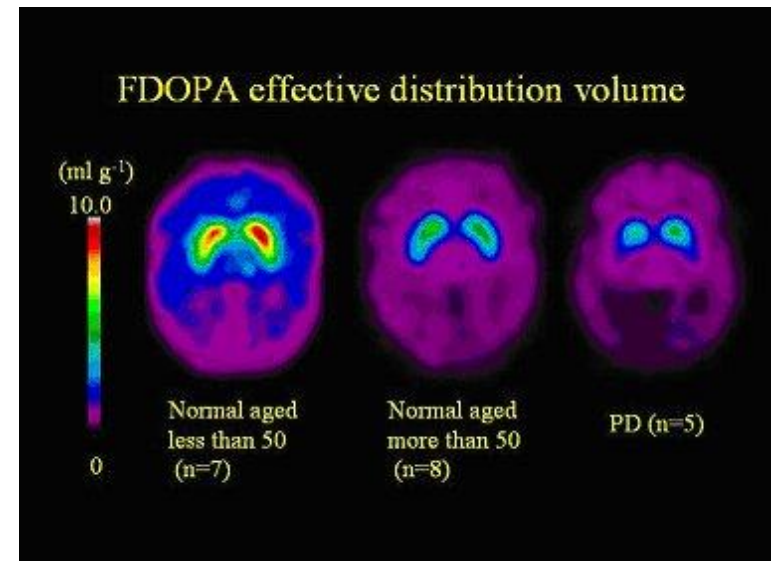
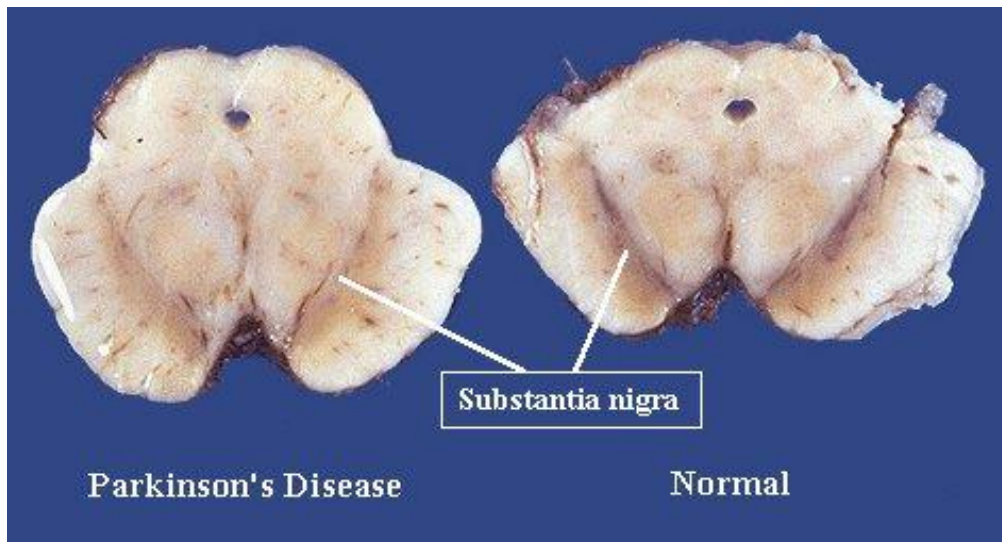
“system degenerations” that involve particular neuronal systems

Parkinson's Disease

Pathology
Substantia Nigra

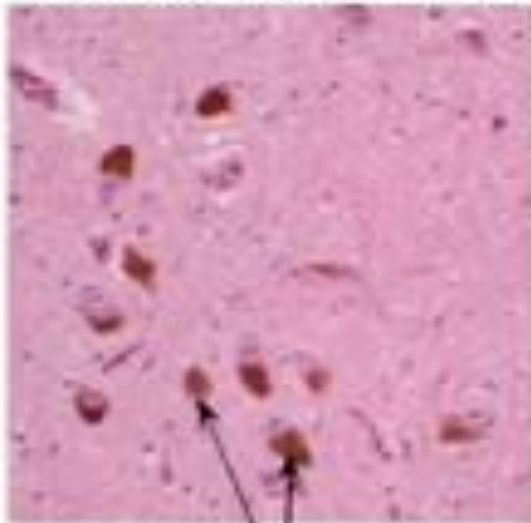


Biochemistry
Striatum



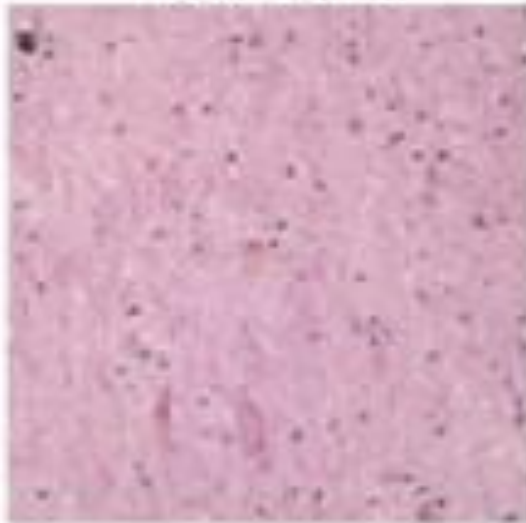
Definition / Pathology

Normal Substantia Nigra

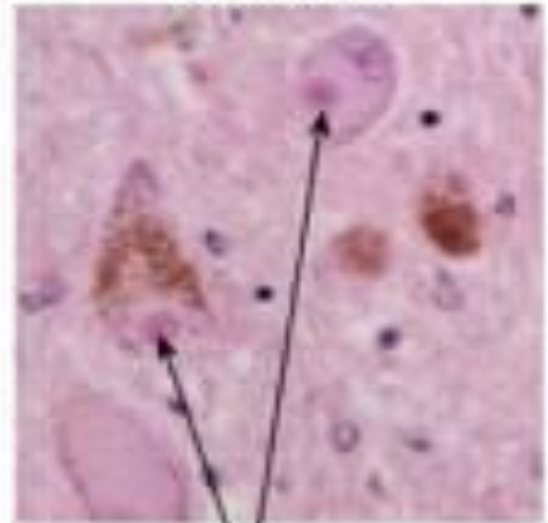


Pigmented neurons

Parkinson's



Parkinson's



Lewy bodies

Motor Neuron Disease

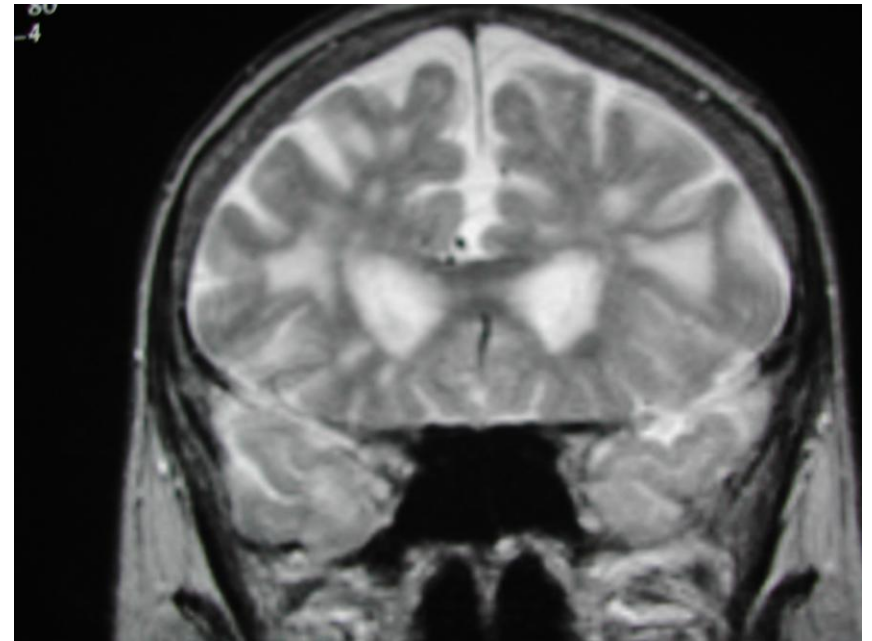
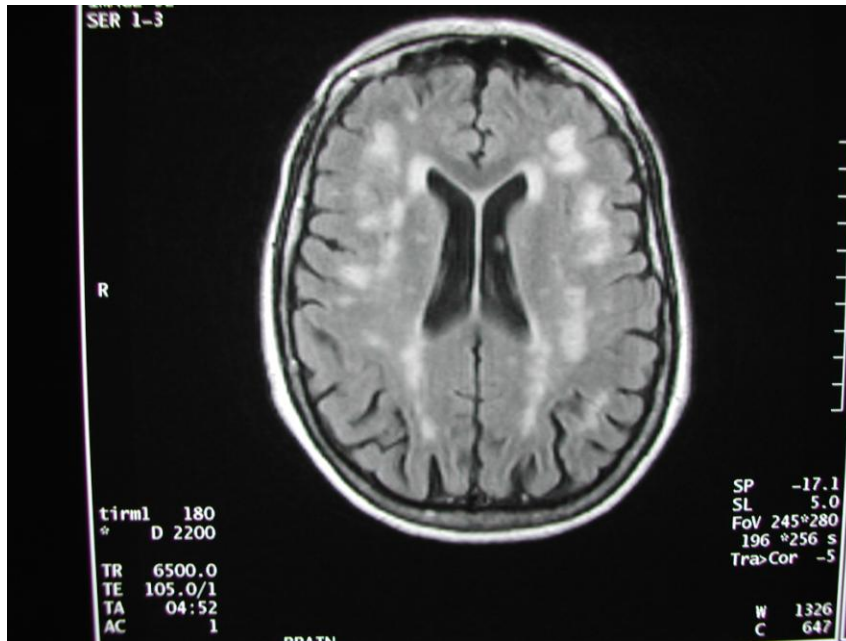
Amyotrophic Lateral Sclerosis (ALS)



- Progressive wasting disease
- Loss of anterior horn motor neurons
- Respiratory failure – VC determines outcome
- Peak onset in 6th decade

J-M. Charcot 1874

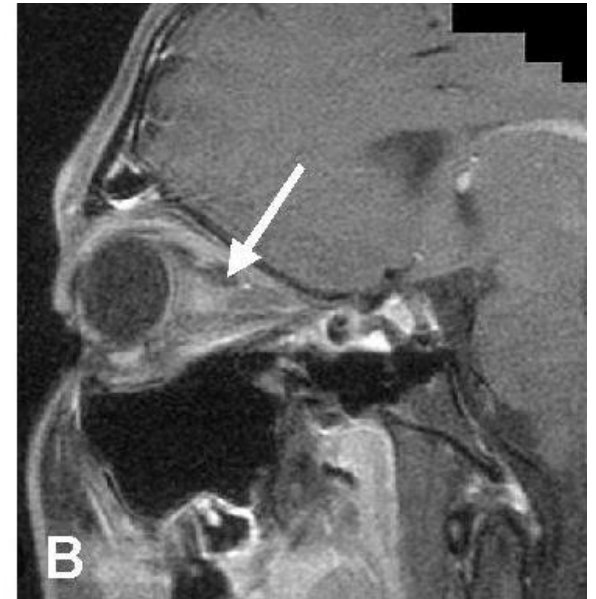
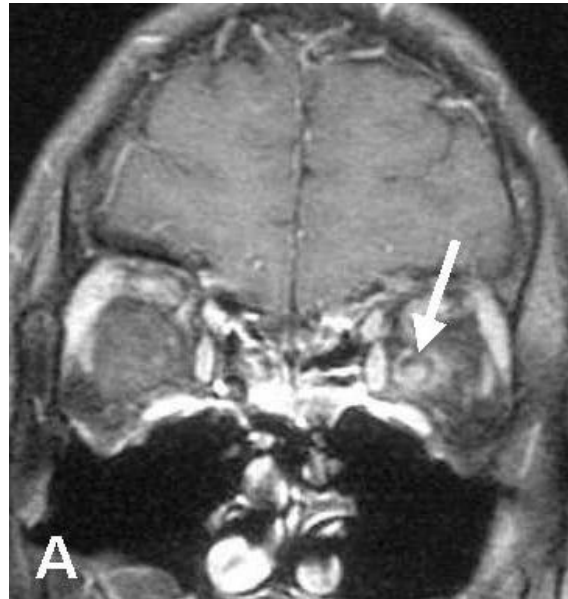
Multiple sclerosis





Optic Neuritis

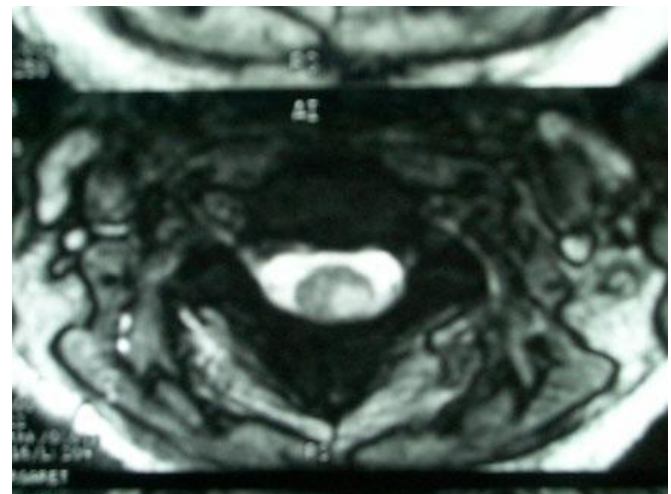
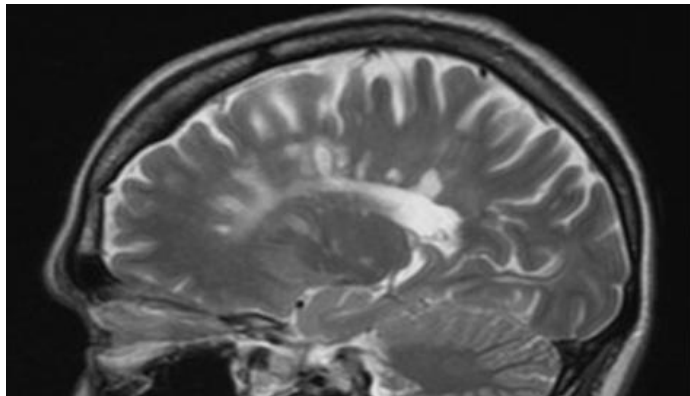
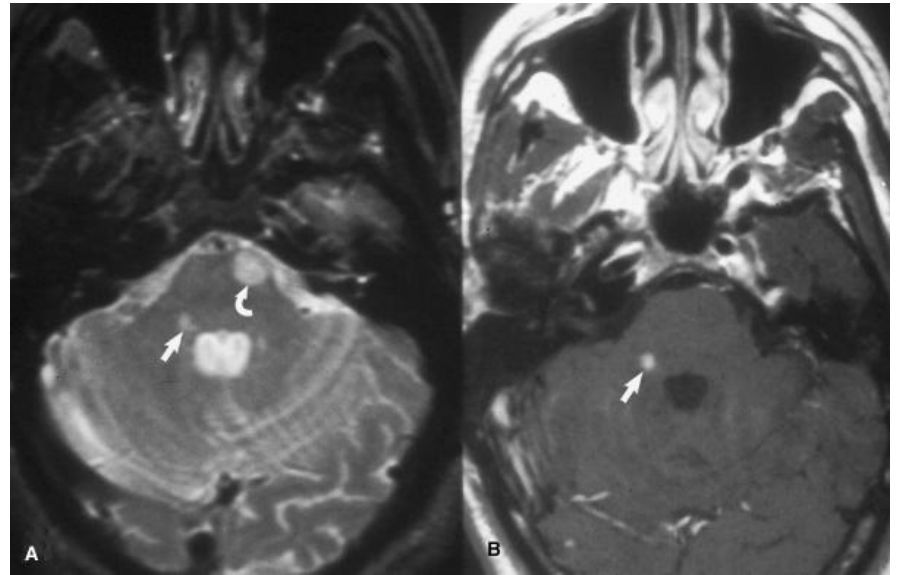
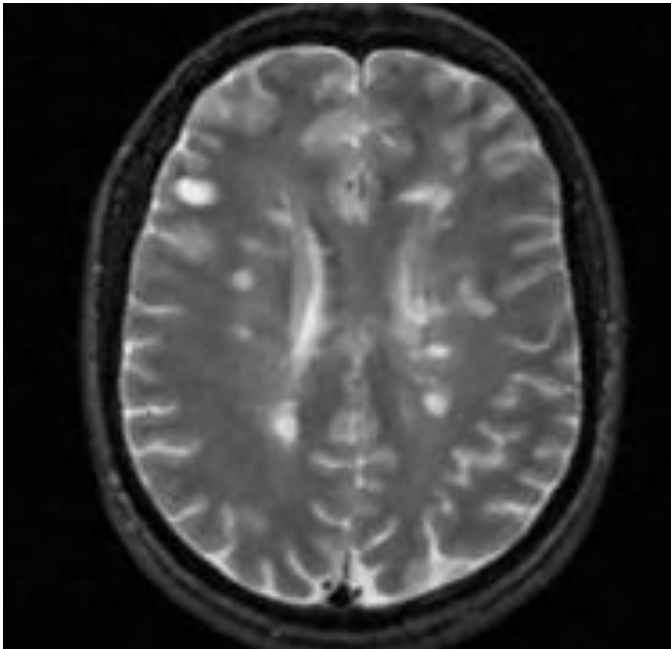
Sometimes asymptomatic



Brainstem Plaques in Multiple Sclerosis

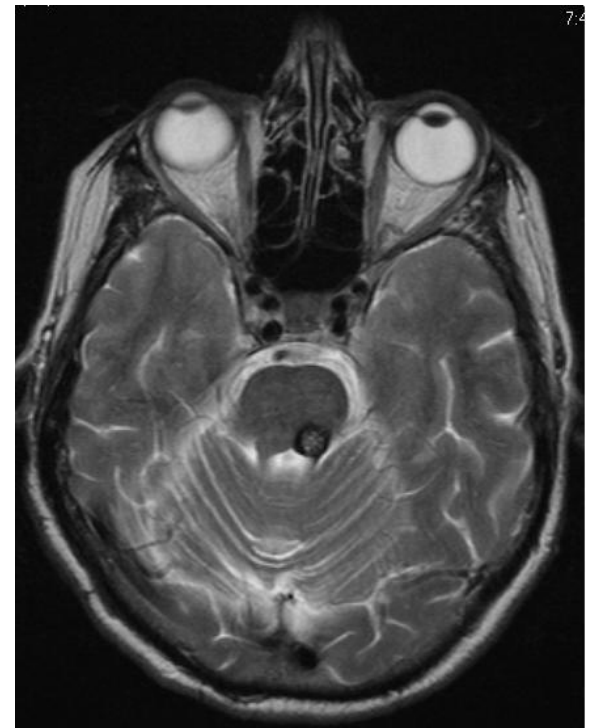
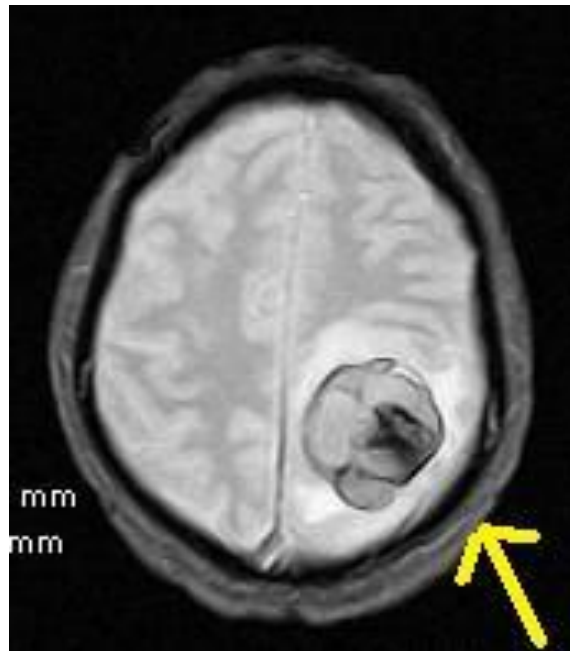
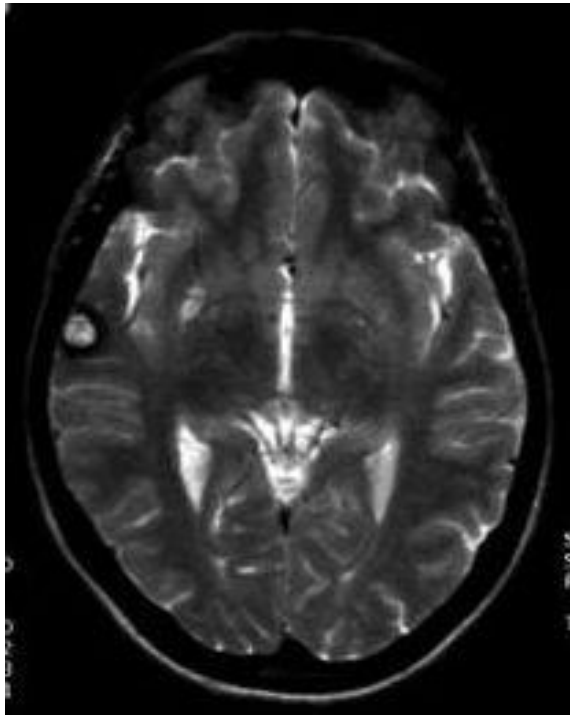


MRI in the Diagnosis of Multiple Sclerosis



Cavernoma

Size & Location Important

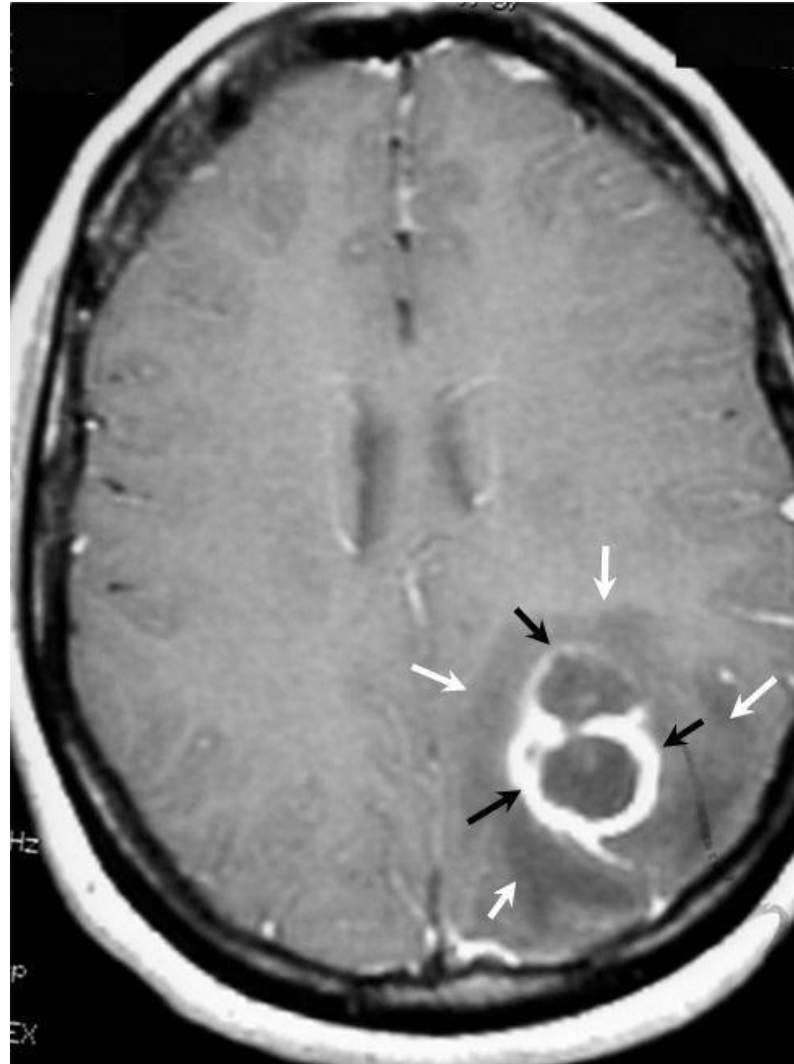


Pilocytic Astrocytoma

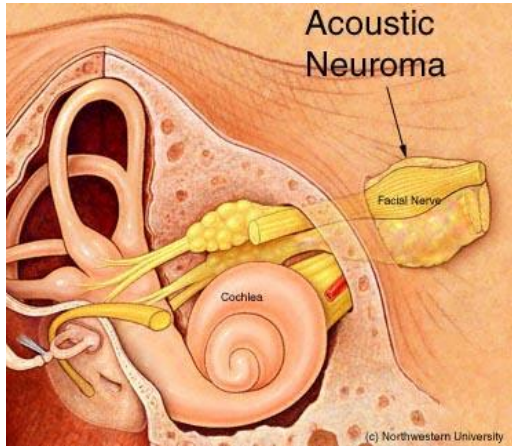
Left Cerebellar Hemisphere



Left Occipital Lobe Abscess



Acoustic Neuroma



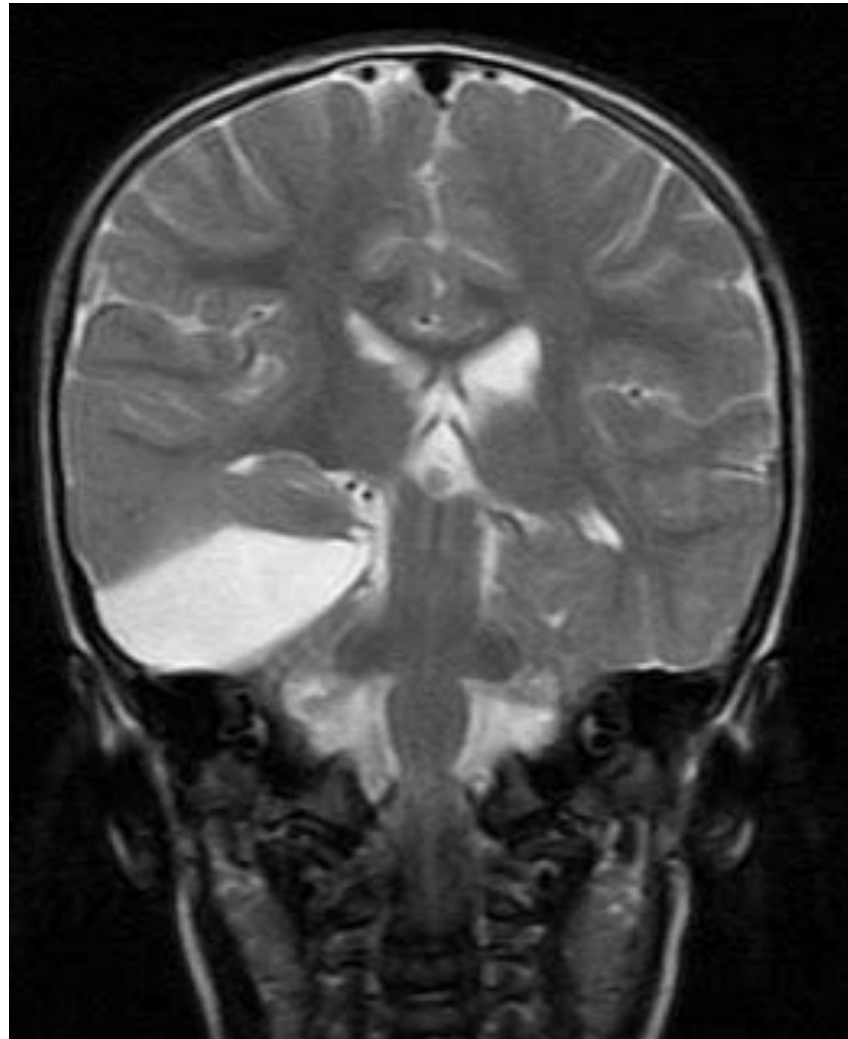
- Vertigo unusual
- Unsteadiness more common
- Tinnitus, or hearing loss
- Facial weakness sometimes
- Loss of ipsilateral corneal reflex



Arachnoid Cyst

Left Middle Cranial Fossa

No Symptoms

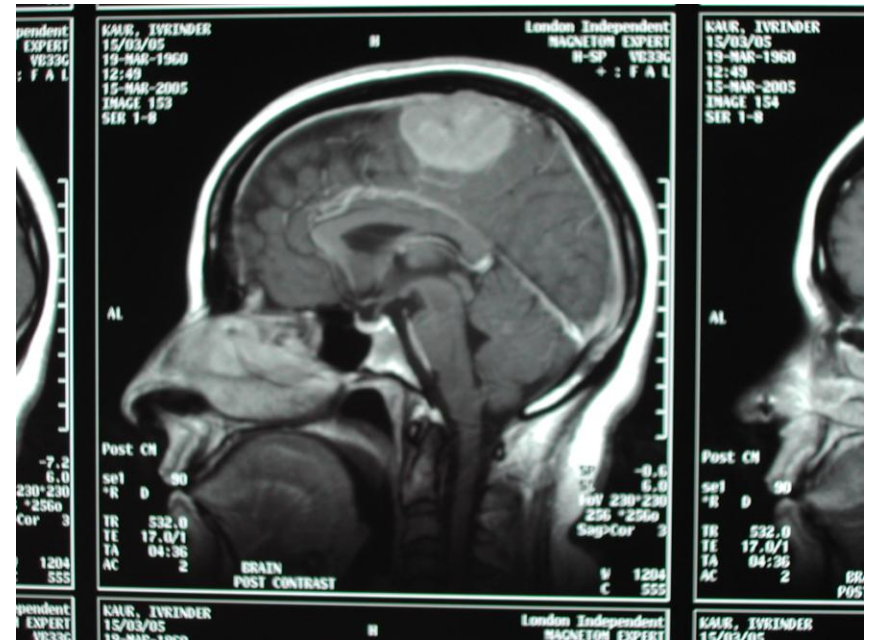
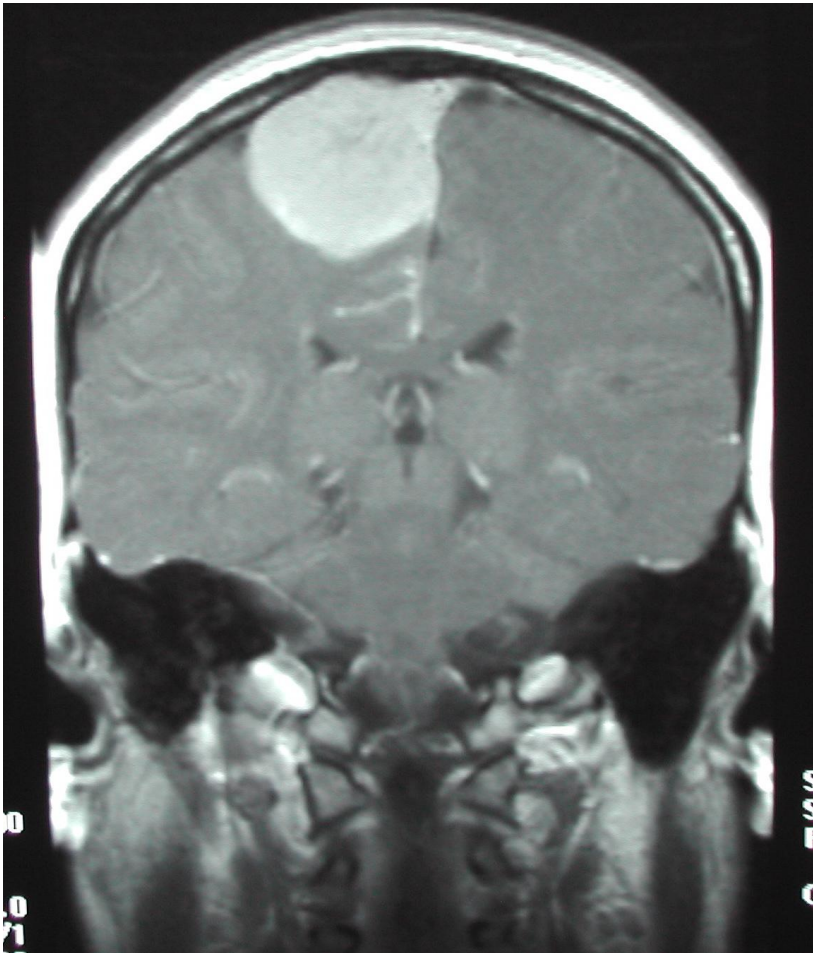


4 Types of Brain Haemorrhage

Following Head Injury



meningioma



Pituitary tumour

